

TTS Reports

V1.0 Stable Build Project Scope Document

Scope, feature inventory, limitations, roadmap, and development log

Document purpose

This document describes what the current TTS Reports application is for, what the V1.0 stable build can do, what it cannot currently do, the intended MVP boundary, the planned roadmap, and the development history of the build.

Item	Details
Application name	TTS Reports / TTS PRO Report Generator
Build reviewed	TTS_REPORTS_V1_0_STABLE_BUILD_VERSION_LABEL.zip
Current version label	TTS Reports V1.0 MVP / V1.0 Stable Build
Technology stack	Python, Flask, ReportLab, PyMuPDF, browser-based HTML/CSS/JavaScript
Primary output	Customer-ready PDF reports generated from imported tester project files
Primary users	Metrel aPAT / ES Manager users, test and tag contractors, support staff, and internal TTS users

1. Executive summary

TTS Reports is a lightweight web-based reporting application designed to make Metrel test and tag project data easier to turn into customer-ready reports. The current V1.0 stable build focuses on a practical MVP: import a supported project file, normalise the data into a clean asset register, review the assets, and generate one of several PDF report types.

The app exists because customers frequently experience friction with existing desktop reporting workflows. TTS Reports provides a simpler interface, avoids unnecessary CSV conversion, supports multiple export formats, and allows users to produce cleaner reports from the same project data.

V1.0 MVP position

This build is not intended to replace every feature of PATLink PC, ES Manager, or a full cloud compliance platform. It is a focused reporting layer that proves whether customers value a simpler path from Metrel project data to usable PDF reports.

2. What the app is for

- Generate clean customer-facing PDF reports from Metrel aPAT and ES Manager project exports.
- Reduce support calls caused by confusing report generation workflows in existing software.
- Let users review imported project data in an Asset Register Dashboard before generating a report.
- Give contractors a faster way to produce Basic, PRO, and Upcoming Retest reports.
- Support tablet-first workflows where users export data from aPAT, access the web app, and generate a report without needing a full PC workflow.
- Provide a future base for a larger reporting product with accounts, cloud storage, branding, customer portals, and recurring compliance reminders.

3. Target users and use cases

User type	Primary need	How TTS Reports helps
Test and tag contractor	Create a clean report for a customer after a job.	Imports the project file, shows latest asset status, and generates a PDF report.
Small business / compliance staff	Review asset status and retest dates without complex reporting software.	Provides a dashboard with status counts, due dates, locations, and search/filter controls.
TTS support team	Help customers produce reports when they struggle with PATLink PC or ES Manager.	Provides a simpler guided workflow and clearer report options.
Internal product / sales team	Demonstrate a value-added customer tool.	Shows potential for customer retention, product education, promotions, and reduced support burden.

4. Current V1.0 workflow

1. Open the TTS Reports web app locally.
2. Go to the Import page.
3. Upload one supported project file: aPAT PDF, aPAT .padfx, aPAT .apx, ES Manager .xlsx, or ES Manager .padfx.
4. Review the imported data on the Home Asset Register Dashboard.
5. Search, filter, or inspect individual assets and test lines.
6. Open the Reports page.
7. Choose Basic Report, PRO Report, or Upcoming Retest Report.
8. Confirm Latest test only customer mode or deliberately open Advanced / Audit options.
9. Preview the report if needed.
10. Generate and download the PDF report.

5. Current feature inventory

Area	Feature	Description
Import	Multi-format import	Supports .pdf, .padfx, .padf, .apx, and .xlsx project files.
Import	Auto-detection	Identifies supported file type from extension and routes it to the relevant parser.
Import	PDF text import	Reads selectable embedded text from the aPAT PDF export. OCR is not used.
Import	PADFX / PADF import	Reads DataSource.padf from the project archive.
Import	APX import	Reads StructureExport/Instruments-style data from the APX archive.
Import	XLSX import	Reads ES Manager spreadsheet workbook XML directly.
Dashboard	Asset Register Dashboard	Displays imported assets as a clean customer-friendly register rather than raw rows.
Dashboard	Summary cards	Shows counts such as total assets, passed, failed, locations, due in 90 days, and overdue.
Dashboard	Search	Searches by asset ID, type, location, user/tester, and test summary.
Dashboard	Status and location filters	Allows narrowing the register by asset status and location.
Dashboard	Column-level filters	Clickable headings focus column filter controls for fields such as Asset ID, Type, Location, Test Date, Retest Date, Tested By, Status, Tests, and Summary.
Dashboard	Table and tree views	Allows a standard table view and an alternate grouped tree-style view.
Dashboard	Asset detail modal	Click an asset row to view a compact detailed test summary.
Reports	Basic Report	Produces a simple title page and results register.
Reports	PRO Report	Produces a cover page, General Data section, results summary, and optional per-appliance detail pages.
Reports	Upcoming Retest Report	Groups assets into Overdue, Due in 30 days, Due in 60 days, and Due in 90 days.
Reports	Report preview	Builds a report preview in an in-app modal/iframe before download.
Reports	Latest test only mode	Locks normal customer reports to the current/latest asset status by default.
Reports	Advanced / Audit mode	Allows date range or full-history reports when deliberately selected.
Settings	Business profile	Stores contractor/business details for pre-filling report fields.
Settings	Report defaults	Stores default report type and standard customer reporting preferences.
Settings	Template defaults	Stores reusable General Data values for report title pages.
Settings	Sign-off defaults	Stores operator/customer sign-off values for report sections.
Settings	Local settings storage	Settings are saved in the browser using localStorage for the customer-facing beta.
Branding	TTS colour palette	Uses TTS Vibrant Red, Seafoam, Charcoal, Beige, and White as the default app/report style.
Build	Windows build support	Includes build_windows.bat for packaging the local desktop-style build with PyInstaller.

Area	Feature	Description
CLI	Command-line report generation	cli.py supports project input, output path, report type, and filter mode.

6. Current pages and interface sections

Page / section	Purpose	Current scope
Home	Asset register dashboard	Dashboard-only view. Report controls have been removed from Home and kept on the Reports page.
Import	Project file upload	Upload a supported project file and populate the internal asset register.
Reports	Report generation	Choose Basic, PRO, or Upcoming Retest; preview and generate PDF output.
Template	Report field editing	Editable General Data/report title page values for report generation.
Settings	Reusable defaults	Business profile, report defaults, template defaults, and sign-off values saved locally.
Help	Guided workflow and explanations	Explains supported files, report types, latest-test mode, file saving, and common fixes.

7. Normalised data model

The app converts different source file formats into a common internal asset register so the dashboard and reports can use the same fields regardless of original export type.

Normalised field	Purpose
Appliance ID / Asset ID	Primary asset identifier used for grouping and reporting.
Type	Asset/appliance type or description.
Location	Customer site, department, area, or asset location.
Test Date	Date of the relevant/latest test.
Retest Date	Next due date where available.
Status	Pass/fail or comparable compliance status.
User / Tested By	Operator or tester value where available in the source data.
Test Summary	Compact summary of the main result.
Detailed test lines	Per-test details used for PRO detail pages and asset modal display.

Source format warning

Different Metrel export formats can store the same project data differently. The app normalises key fields, but some exports may omit fields that are present in other formats. The ES Manager .xlsx export may not contain every retest-date detail available in .padfx or .apx files.

8. What the app cannot currently do

Limitation	Explanation
No cloud accounts	The V1.0 stable build does not include user accounts, authentication, multi-user access, or cloud dashboards.
No cloud storage	Imported files and generated PDFs are handled locally in the current build. There is no hosted project library or customer portal.
No OCR	PDF import reads selectable embedded text only.

Limitation	Explanation
	Scanned/image-only PDFs are not supported unless OCR is added later.
No full PATLink / ES Manager replacement	The app generates reports from exported data. It does not manage instruments, upload to testers, synchronise devices, or replace the full vendor ecosystem.
No live aPAT integration	The current workflow relies on exported files rather than direct tablet/app API integration.
No automated email delivery	Generated PDF reports are downloaded manually. Emailing reports is out of scope for V1.0.
No scheduled reminders	Upcoming Retest Report is generated manually. Automatic email/SMS reminders are planned but not present.
No stored report history	The interface notes that report history is not yet stored in this local build.
No advanced compliance engine	The app formats and filters imported data. It does not independently certify legal compliance or replace professional judgement.
No payment/subscription layer	The stable build has no billing, licence enforcement, subscriptions, trials, or product tiers.
No built-in customer CRM	Customer information can prefill reports but is not yet a managed customer database.
Limited mobile polish	The app can support tablet-oriented workflows, but V1.0 should still be tested across tablets, browsers, and screen sizes before customer rollout.

9. MVP scope boundary

In scope for V1.0	Out of scope for V1.0
Import supported Metrel export files.	Direct tester synchronisation or instrument management.
Normalise asset data for reporting.	Replacing vendor software entirely.
Home Asset Register dashboard.	Cloud asset database and customer portals.
Basic, PRO, and Upcoming Retest PDFs.	Automated recurring reminder system.
Latest-test default customer mode.	Complex compliance rule engine.
Advanced date range/full-history audit mode.	Built-in billing, subscription tiers, or licence keys.
Local settings and report defaults.	Enterprise user management, roles, and permissions.
Local app packaging and simple Windows build.	Production cloud hosting architecture.

10. Technical architecture overview

Component	Current implementation
Backend framework	Flask application in app.py.
Report generation engine	ReportLab PDF generation in report_core.py.
PDF parsing dependency	PyMuPDF for reading selectable text from aPAT PDF exports.
Frontend	Single HTML template with CSS and JavaScript for navigation, dashboard, filters, settings, modals, and report controls.
Storage	Uploaded files and PDFs are saved to local uploads/output folders during use. Settings use browser localStorage.
Session state	The last imported project file path is retained in Flask session so reports can generate after preview/import even if the browser clears the file input.
CLI	Command-line interface for generating reports outside the web UI.
Packaging	build_windows.bat supports creating a Windows executable build with PyInstaller.

11. Data and privacy considerations

- V1.0 is local-first. Imported customer project files and generated PDF reports stay on the machine running the app unless the user manually moves or shares them.
- The current build does not upload data to a cloud service by default.
- Browser localStorage is used for saved settings, so settings are tied to the browser/device and should not be treated as secure enterprise storage.
- If the app becomes hosted, the future product will need authentication, secure file handling, retention rules, data deletion, backups, audit logs, and privacy policy updates.
- Imported files can contain customer, site, equipment, and operator information. Any hosted version must treat them as sensitive business records.

12. Planned features and roadmap

Stage	Feature	Description
V1.0 hardening	Importer regression tests	Keep sample files for every supported import type and verify consistent asset counts, statuses, dates, and report output after each build.
V1.0 hardening	PDF import robustness	Improve handling of aPAT PDF layout variations and provide clearer errors when PDF text is not selectable.
V1.0 hardening	Report QA suite	Automated checks for Basic, PRO, and Retest report generation after each change.
V1.0 hardening	Responsive UI pass	Confirm dashboard, reports page, and settings work cleanly on common laptop/tablet screen sizes.
V1.1	Report history	Store generated report records locally with filename, date, customer, report type, and source file.
V1.1	Customer/project details manager	Save reusable customer profiles, site details, and project metadata.
V1.1	Branding controls	Allow logo upload, brand colour overrides, footer text, and report header settings.
V1.1	Import validation summary	Show warnings for missing retest dates, missing asset IDs, unsupported rows, and source differences.
V1.1	Export asset register	Export dashboard view to CSV/XLSX for support and audit use.
V1.2	Email-ready report pack	Generate report plus cover email text, customer summary, and retest action list.
V1.2	Marketing/news panel	Non-intrusive promotional area for specials, consumables, product news, support updates, or training links.
V1.2	Template library	Saved report templates for different customer types or service styles.
V2.0	Hosted web app	User accounts, login, cloud project storage, secure report library, and device-independent access.
V2.0	Customer portals	Customers can access reports, asset registers, retest status, and downloadable documents.
V2.0	Automated retest reminders	Scheduled email/SMS reminders for due and overdue assets.
V2.0	Subscription/licensing	Paid plans, trials, account limits, usage tracking, and support tiers.
V2.0+	Advanced analytics	Compliance trends, failure rates, technician activity, asset lifecycle, and revenue opportunities.
V2.0+	Integrations	Possible integrations with Shopify

Stage	Feature	Description
		support pages, CRM, email platforms, accounting, or external compliance systems.

13. Development log

Phase	Change	Notes
Initial concept	Problem identified	Customers struggle to generate usable reports from existing Metrel reporting software such as PATLink PC and ES Manager.
Early prototype	CSV-based reporting	Initial report generator explored CSV inputs and produced Basic/PRO-style PDF outputs.
PADFX direction	Raw .padfx support	Shifted from requiring CSV conversion to importing the raw .padfx project export directly.
Dashboard build	Web app interface	Home page evolved into a functional web-app dashboard with imported data shown in a table rather than a raw upload form.
Filtering update	Latest test only	Normal customer reports were locked to latest test only to prevent older repeated tests appearing in customer-facing reports.
Advanced reporting	Date range and all-history audit mode	Historical reporting was moved into an Advanced / Audit section for deliberate internal/audit use.
Report set expansion	Basic, PRO, and Upcoming Retest	Reports dashboard gained clear separate actions for Basic Report, PRO Report, and Upcoming Retest Report.
Settings expansion	Business profile and defaults	Settings were expanded to include business profile, report defaults, template defaults, sign-off, import, output, privacy, and developer areas.
Multi-format import	.padfx, .apx, .xlsx support	The app was updated so customers can import project data from direct aPAT and ES Manager export formats.
PDF import addition	aPAT PDF import	Selectable-text PDF import was added to reduce dependency on proprietary project formats and improve customer flexibility.
Output consistency fix	Normalisation across formats	Importer logic was corrected so equivalent project data from different file types produces consistent report results where the source data allows.
Report generation fix	Hidden required settings issue	Resolved/avoided browser validation issues caused by hidden required settings fields blocking report generation.
Stable V1 workflow	Home dashboard / Reports separation	Report controls were removed from Home. Home now focuses on the Asset Register dashboard, while report generation occurs from the Reports page only.
Stable V1 filter update	Column-level dashboard filtering	Asset Register Dashboard gained clickable column headings and filters, plus a Tested By column using imported tester/operator values.
V1.0 stable label	Stable build packaged	Current stable zip reviewed as the basis for this project scope document.

14. Recommended acceptance criteria for V1.0

- A user can import each supported sample file type without errors.
- Equivalent sample project data produces matching asset counts and key results across supported formats where source fields are present.
- Home shows only the Asset Register dashboard and does not expose report generation controls.
- Reports can only be generated from the Reports page workflow.
- Basic Report, PRO Report, and Upcoming Retest Report generate valid downloadable PDFs.
- Latest test only remains the default customer reporting mode.
- Advanced / Audit reporting is visually separated and clearly labelled.
- Settings can be saved, reset, and used to prefill report data without blocking report generation.
- Generated reports are readable, branded, and suitable for a customer-facing beta.
- The Help page accurately explains supported files, workflow, report types, and common fixes.

15. Key risks and recommendations

Risk	Recommendation
Different export formats may contain different fields.	Keep a visible import validation summary and document expected source differences.
PDF export layouts may change between aPAT versions.	Add regression samples for every known aPAT PDF layout and fail gracefully when parsing is uncertain.
Local settings can be lost or browser-specific.	For MVP this is acceptable; for hosted use, migrate settings to authenticated user storage.
Users may expect the app to be a full compliance platform.	Position V1.0 clearly as a report generator and asset register viewer, not as a replacement for all vendor software.
Support staff need confidence in results.	Create a fixed QA pack with source files and expected PDF outputs for every release.
Report UI can become cluttered as features grow.	Keep Home, Import, Reports, Template, Settings, and Help separated by workflow purpose.

16. Suggested product positioning

TTS Reports is best positioned as a customer support and productivity tool: a simpler way for Metrel users to turn exported test data into readable reports without fighting complex reporting software. The MVP should be promoted as a fast reporting layer first, then expanded only after real users confirm which workflows matter most.

Positioning sentence

Import your Metrel project file, review your asset register, and generate a clean customer report in minutes - without wrestling with the old reporting workflow.

Appendix A - Supported file behaviour

File type	Current behaviour	Important limitation
.pdf	Reads selectable embedded text from aPAT PDF reports.	Image-only/scanned PDFs are not supported without OCR.
.padfx / .padf	Reads project archive data and extracts key asset/test fields.	Archive structure must match expected export format.
.apx	Reads APX export data from JSON-style archive files.	Field availability depends on APX export contents.
.xlsx	Reads ES Manager workbook XML sheets.	May not include every retest-date or detail field available in project archives.